

Multijet balance: smearing the recoil in MC

Four configurations, $R = 0.4$

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Common setup

- ▶ $x_j = p_{T,\text{lead}}/|\vec{p}_{T,2} + \vec{p}_{T,3}|$; the recoil is $|\vec{p}_{T,2} + \vec{p}_{T,3}|$.
- ▶ Reprocessed data and Pythia8 trees (new skim, per-event seeded JER smearing).
- ▶ MC: Jet12, Jet20, Jet30; Jet12 starts at truth $p_T = 0$.
- ▶ Leading reco jet capped at 35 GeV in Jet12 and 50 GeV in Jet20.
- ▶ Leading jet ≥ 20 GeV, jets 2 and 3 ≥ 7 GeV each, $|\eta| < 0.7$, $\Delta\phi_{12} > 3\pi/4$, $\Delta\phi_{13} > \pi/2$.
- ▶ Leading- p_T bins 20–25, 25–30, 30–35, 35–50 GeV.
- ▶ MC reweighted in leading p_T and z vertex; two passes.
- ▶ Only the treatment of the MC recoil changes between the configurations.

The four configurations

	jets 2 and 3	recoil smearing	recoil cut
1	each smeared on its own	none	none
2	each smeared on its own	none	≥ 14 GeV
3	added unsmeared	sum smeared once, every event	≥ 14 GeV
4	2 truth jets: added unsmeared otherwise: each smeared on its own	2 truth jets: sum smeared once otherwise: none	≥ 14 GeV

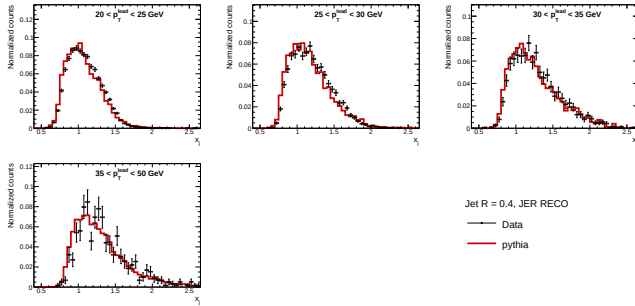
- ▶ In configuration 3 the smearing width is taken at the unsmeared recoil p_T , so truth is used only for the sample stitching.
- ▶ Configuration 4 smears the recoil only in events with exactly two truth jets ≥ 7 GeV, with the width at the recoil truth jet's p_T .
- ▶ Its other events (about 30% at $R = 0.4$) smear jets 2 and 3 individually, as in configuration 2.
- ▶ The 7 GeV jet cuts use the values before any recoil smearing.

Result: data/MC of $\langle x_j \rangle$, $R = 0.4$

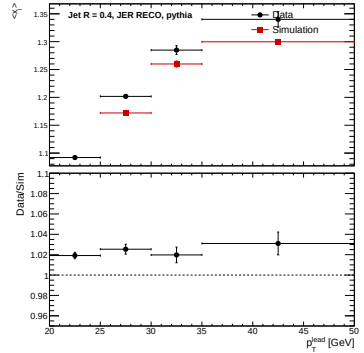
	20–25	25–30	30–35	35–50
1: individual, no recoil cut	1.019 ± 0.003	1.025 ± 0.005	1.020 ± 0.008	1.031 ± 0.011
2: individual, recoil cut	1.026 ± 0.003	1.031 ± 0.005	1.020 ± 0.008	1.030 ± 0.011
3: recoil smeared, all events	0.998 ± 0.004	0.990 ± 0.005	0.979 ± 0.007	0.980 ± 0.011
4: recoil smeared, 2 truth jets	0.997 ± 0.004	1.001 ± 0.005	0.985 ± 0.008	0.986 ± 0.011
hybrid: all events, truth width if 2	1.000 ± 0.004	0.991 ± 0.005	0.979 ± 0.007	0.980 ± 0.011
data events	16090	5346	1640	590

- ▶ The recoil cut removes about 1.5% of data events and moves the ratio by at most 0.007.
- ▶ Smearing the summed recoil once raises the MC $\langle x_j \rangle$ by 2.5–5%.
- ▶ Hybrid (configuration 3 with the truth width in two-truth-jet events) matches configuration 3, so the width choice does not matter.

Configuration 1: individual smearing, no recoil cut

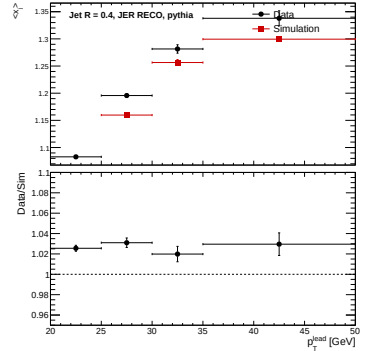
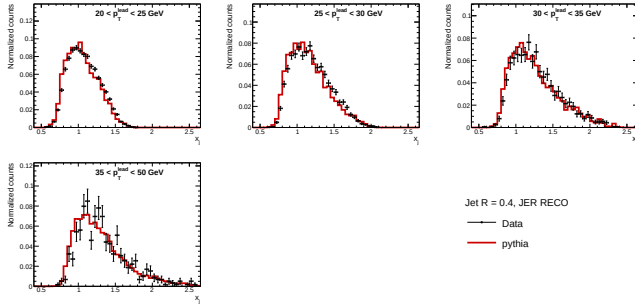


Jet R = 0.4, JER RECO
— Data
— pythia



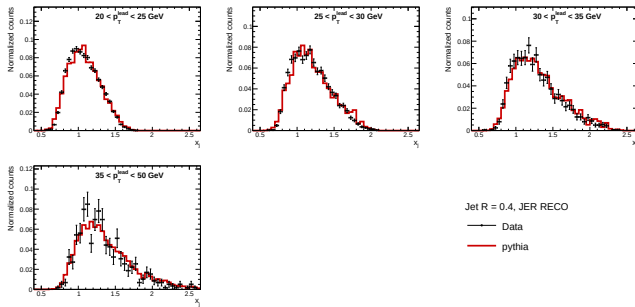
- ▶ Data lies 2–3% above MC in every bin.
- ▶ The MC x_j distribution peaks slightly lower than data.

Configuration 2: individual smearing, recoil ≥ 14 GeV



- ▶ Nearly identical to configuration 1.
- ▶ Two jets of at least 7 GeV almost always give a recoil above 14 GeV.

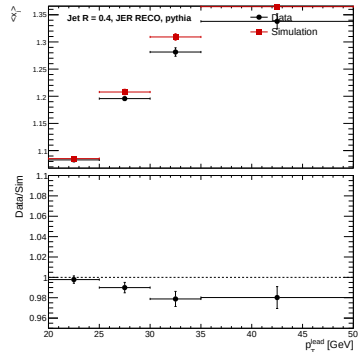
Configuration 3: add unsmeared, smear the recoil, recoil ≥ 14 GeV



Jet R = 0.4, JER RECO

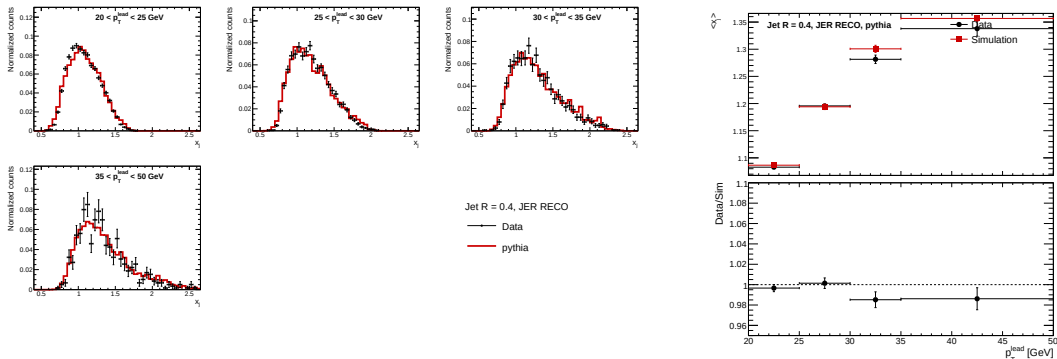
— Data

— pythia



- ▶ The distributions agree over the whole x_j range in the first three bins.
- ▶ The ratio is about 1.00 at 20–25 GeV and 0.98–0.99 above.

Configuration 4: recoil smearing only in two-truth-jet events



- Events with exactly two truth jets ≥ 7 GeV: jets 2 and 3 added unsmeared, the sum smeared at the recoil truth jet's p_T .
- All other events: jets 2 and 3 smeared individually.
- The ratio is 1.00 at 20–30 GeV and about 0.985 above.

Summary

- ▶ How the recoil is smeared, not the recoil cut, sets the data/MC agreement.
- ▶ Smearing jets 2 and 3 individually leaves a 2–3% data excess.
- ▶ Adding jets 2 and 3 unsmeared and smearing the sum once removes it.
- ▶ Configuration 3 agrees with configuration 4 at 20–25 GeV and is about 1% lower above.
- ▶ That difference comes from the events without exactly two truth jets, which configuration 4 smears individually; the width choice plays no role.
- ▶ Configuration 3 needs no truth information, so it is the simpler definition to document.